

DIGITAL MIRROR

AI-Powered Multimodal Behavioral Analysis & Simulation Framework

1. Problem Statement

Modern social media platforms are designed to maximize user engagement using algorithm-driven recommendation systems, short-form videos, emotionally triggering content, and addictive visual patterns. Continuous exposure to such content gradually affects human behavior, emotional stability, cognitive focus, sleep patterns, communication habits, and digital dependency. Existing digital wellbeing systems mainly focus on screen-time monitoring and application blocking. However, they fail to computationally analyze the deeper psychological and behavioral influence created by algorithmic content exposure. DIGITAL MIRROR is proposed as a research-oriented AI-powered multimodal behavioral analysis platform that analyzes text, audio, image, and video inputs to identify behavioral addiction patterns, emotional instability, attention fragmentation, cognitive fatigue, and algorithmic influence. The system introduces computational behavioral indicators such as Dopamine Trigger Score, Attention Fragmentation Index, Sleep Disruption Score, Emotional Dependency Score, and Digital Behavioral Risk Index (DBRI) to model the psychological impact of digital exposure.

2. Proposed Solution

The proposed solution uses Artificial Intelligence, Behavioral Psychology Modeling, NLP, Computer Vision, and Multimodal Analysis techniques to computationally evaluate the psychological and behavioral impact caused by social media algorithms. The system accepts multimodal inputs such as: • Text • Audio • Images • Videos The platform performs: • Sentiment Analysis • Emotional Analysis • Addiction Pattern Detection • Attention Fragmentation Analysis • Sleep Disruption Analysis • Behavioral Risk Prediction The system calculates research-oriented behavioral metrics including: • Dopamine Trigger Score • Emotional Dependency Score • Cognitive Fatigue Score • Algorithmic Influence Score • Digital Behavioral Risk Index (DBRI) Finally, the platform generates: • Behavioral Insights • Psychological Risk Reports • Emotional Analytics • Future Behavioral Simulations • Visualization Dashboards

3. Workflow

Step	Process
1	User uploads text, audio, image, and video inputs
2	System preprocesses and cleans the input data
3	AI modules analyze emotions, addiction patterns, and behavior
4	Behavioral psychology metrics are calculated
5	Behavioral Risk Engine generates risk scores
6	Digital Behavioral Risk Index (DBRI) is generated
7	Dashboard visualizes behavioral insights and analytics
8	Future behavioral simulation and AI ethics awareness are generated

4. Technologies Used

- Python
- Machine Learning
- NLP (Natural Language Processing)
- Computer Vision
- Speech Emotion Recognition
- OpenCV
- Pandas & NumPy
- Streamlit / Flask
- Data Visualization

5. Project Vision

DIGITAL MIRROR is designed to help users understand how algorithm-driven social media platforms gradually influence human psychology, emotional behavior, cognitive focus, and digital dependency using computational behavioral analysis and AI ethics awareness.